

Q9. Program to check for leap year?

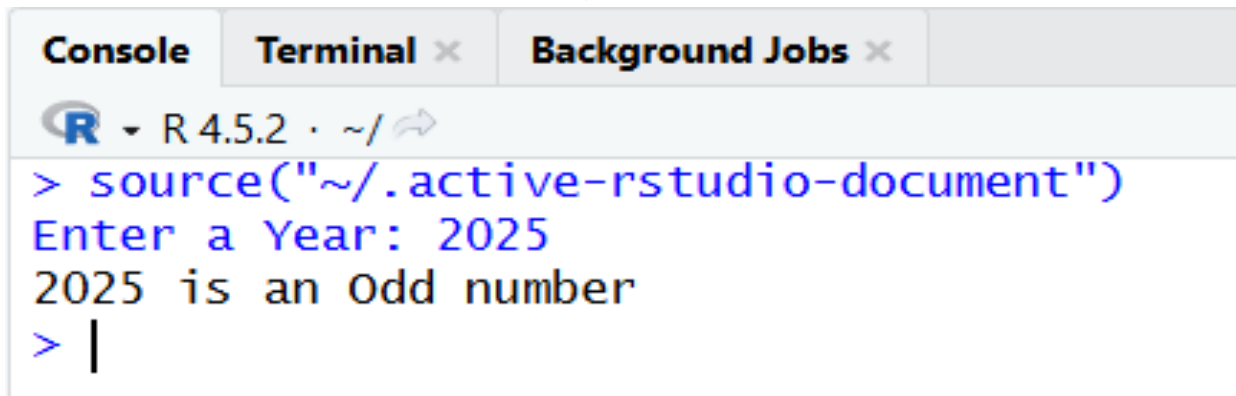
```
# Taking input from user  
year <- as.integer(readline(prompt = "Enter a year: "))  
  
# Checking leap year  
if ((year %% 4 == 0 && year %% 100 != 0) || (year %% 400 == 0)) {  
  cat(year, "is a Leap Year")  
} else {  
  cat(year, "is not a Leap Year")  
}
```

Output:

Console	Terminal ×	Background Jobs ×
 R 4.5.2 · ~/ ↩ > source("~/active-rstudio-document") Enter a year: 2024 2024 is a Leap Year >		

Q10. Check whether number is odd or even?

```
# Taking input from user
num <- as.integer(readline(prompt = "Enter a Year: "))
# Checking odd or even
if (num %% 2 == 0) {
  cat(num, "is an Even number")
} else {
  cat(num, "is an Odd number")
}
```

Output:

```
Console Terminal x Background Jobs x
R • R 4.5.2 • ~/
> source("~/active-rstudio-document")
Enter a Year: 2025
2025 is an Odd number
> |
```

Q11. Check if a number is positive negative or zero?

```
# Taking input from user

num <- as.numeric(readline(prompt = "Enter a number: "))

# Checking the number

if (num > 0) {

  cat(num, "is a Positive number")

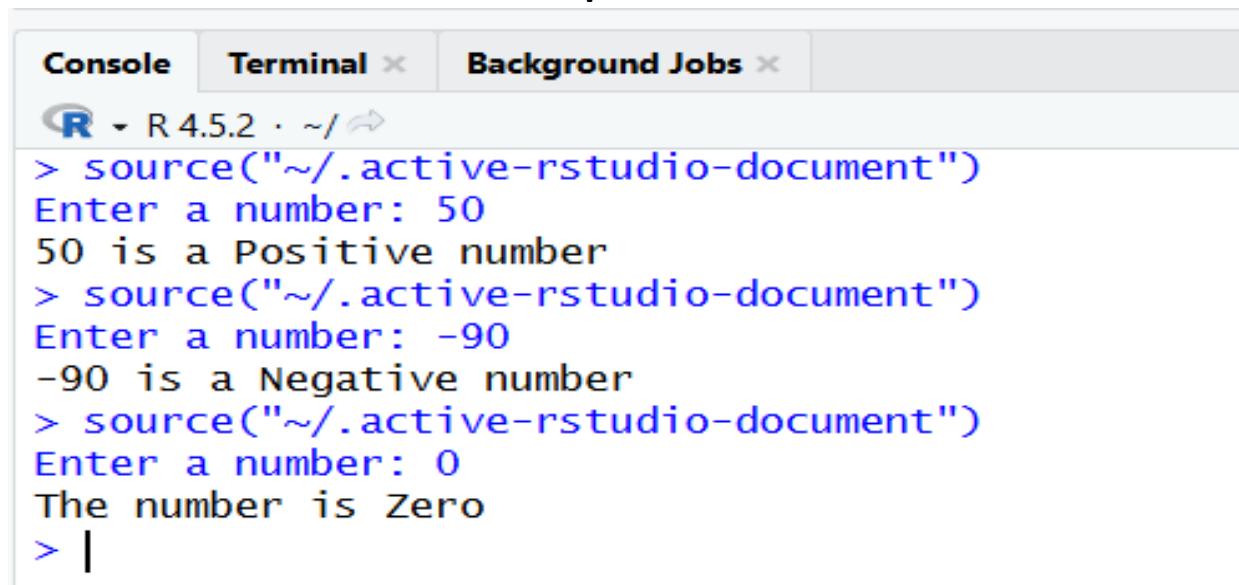
} else if (num < 0) {

  cat(num, "is a Negative number")

} else {

  cat("The number is Zero")

}
```

Output:A screenshot of the R Studio console window. The window has three tabs: 'Console', 'Terminal', and 'Background Jobs'. The 'Console' tab is active. The console shows the R prompt '>' followed by the command 'source("~/active-rstudio-document")' which is executed three times. Each time, it prompts the user to 'Enter a number:'. The first input is '50', resulting in the output '50 is a Positive number'. The second input is '-90', resulting in the output '-90 is a Negative number'. The third input is '0', resulting in the output 'The number is Zero'. The console ends with a prompt '> |'.

Q12. Find the sum of Natural Numbers?

```
# Taking input from user

n <- as.integer(readline(prompt = "Enter a natural number: "))

# Finding sum of first n natural numbers

sum <- 0

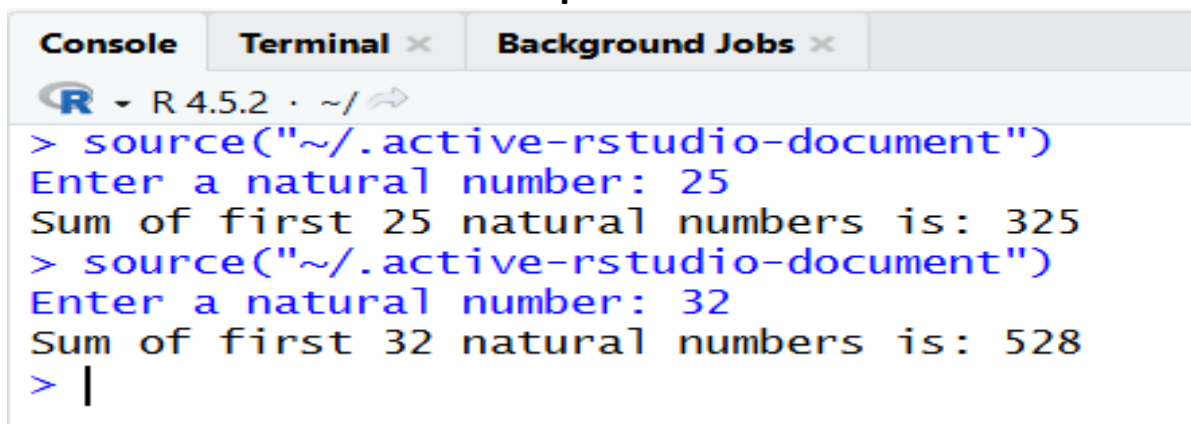
for (i in 1:n) {

  sum <- sum + i

}

# Displaying result

cat("Sum of first", n, "natural numbers is:", sum)
```

Output:

The screenshot shows the R console interface with three tabs: Console, Terminal, and Background Jobs. The Console tab is active, displaying the R prompt and the output of the script. The script is executed twice, first with input 25 and then with input 32. The output for 25 is 'Sum of first 25 natural numbers is: 325' and for 32 is 'Sum of first 32 natural numbers is: 528'. The console title bar indicates 'R 4.5.2 · ~/'. The prompt character is '>'.

```
R 4.5.2 · ~/
> source("~/active-rstudio-document")
Enter a natural number: 25
Sum of first 25 natural numbers is: 325
> source("~/active-rstudio-document")
Enter a natural number: 32
Sum of first 32 natural numbers is: 528
> |
```

Q13. Convert Decimal into Binary using Recursion?

```
# Recursive function to convert decimal to binary

decimalToBinary <- function(n) {

  if (n == 0) {

    return("")

  } else {

    return(paste0(decimalToBinary(n %/% 2), n %% 2))

  }

}

# Taking input from user

num <- as.integer(readline(prompt = "Enter a decimal number: "))

if (num == 0) {

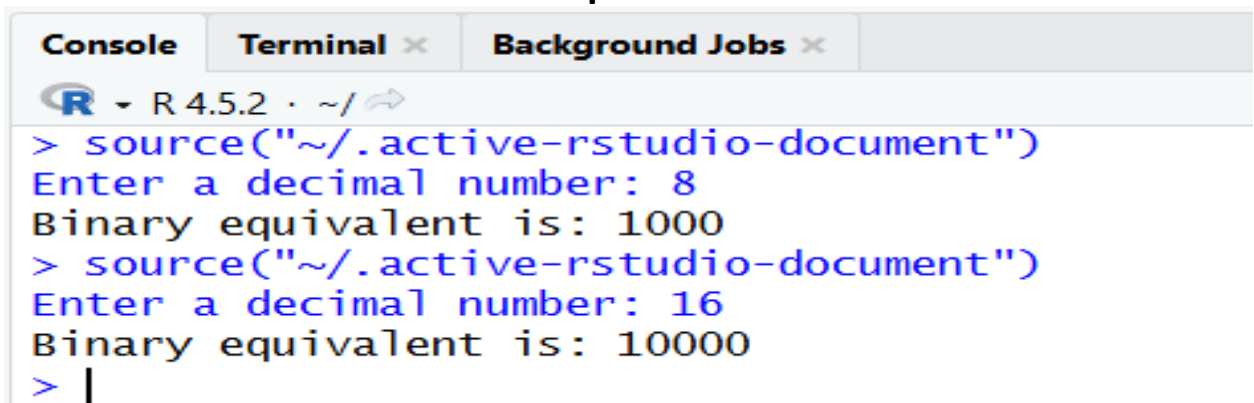
  cat("Binary equivalent is: 0")

} else {

  binary <- decimalToBinary(num)

  cat("Binary equivalent is:", binary)

}
```

Output:The screenshot shows the RStudio interface with the Console pane active. The title bar indicates 'R 4.5.2 · ~/'. The console output shows two successful conversions: 8 to 1000 and 16 to 10000. The prompt 'Enter a decimal number:' is shown in blue text, and the output 'Binary equivalent is:' is in black text. The prompt character '>' is also in blue.

```
R 4.5.2 · ~/
> source("~/active-rstudio-document")
Enter a decimal number: 8
Binary equivalent is: 1000
> source("~/active-rstudio-document")
Enter a decimal number: 16
Binary equivalent is: 10000
> |
```

Q14. Find the factorial of number using Recursion?

```
# Recursive function to find factorial

factorial_rec <- function(n) {

  if (n == 0 || n == 1) {

    return(1)

  } else {

    return(n * factorial_rec(n - 1))

  }

}

# Taking input from user

num <- as.integer(readline(prompt = "Enter a number: "))

if (num < 0) {

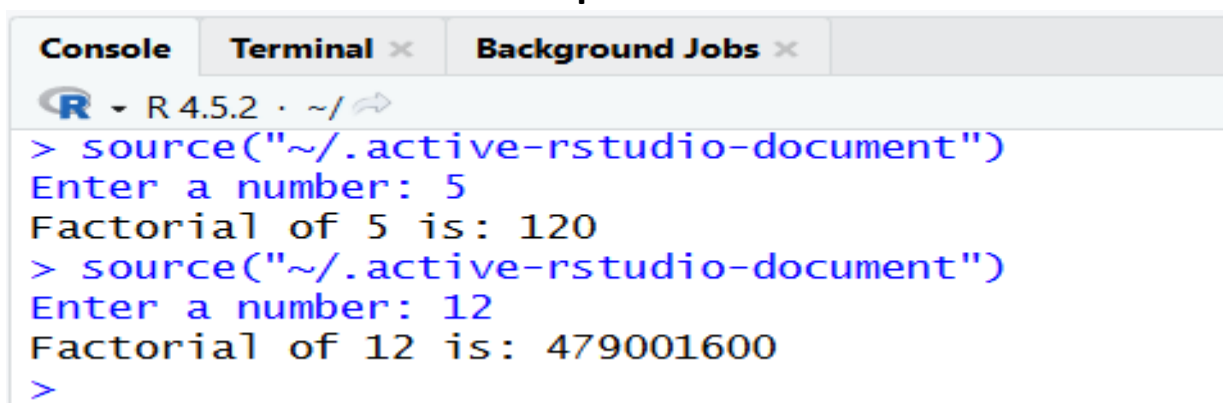
  cat("Factorial is not defined for negative numbers")

} else {

  result <- factorial_rec(num)

  cat("Factorial of", num, "is:", result)

}
```

Output:

The screenshot shows the R console interface with tabs for Console, Terminal, and Background Jobs. The console displays the execution of the R script. It starts with the prompt '> source("~/active-rstudio-document")', followed by the user input '5' and the output 'Factorial of 5 is: 120'. This is repeated for the input '12', resulting in the output 'Factorial of 12 is: 479001600'. The prompt '>' is visible at the bottom of the console.

```
R - R 4.5.2 - ~/
> source("~/active-rstudio-document")
Enter a number: 5
Factorial of 5 is: 120
> source("~/active-rstudio-document")
Enter a number: 12
Factorial of 12 is: 479001600
>
```

Q15. Program to find HCF or GCD of a number?

```
# Recursive function to find GCD

gcd <- function(a, b) {
  if (b == 0) {
    return(a)
  } else {
    return(gcd(b, a %% b))
  }
}

# Taking input from user

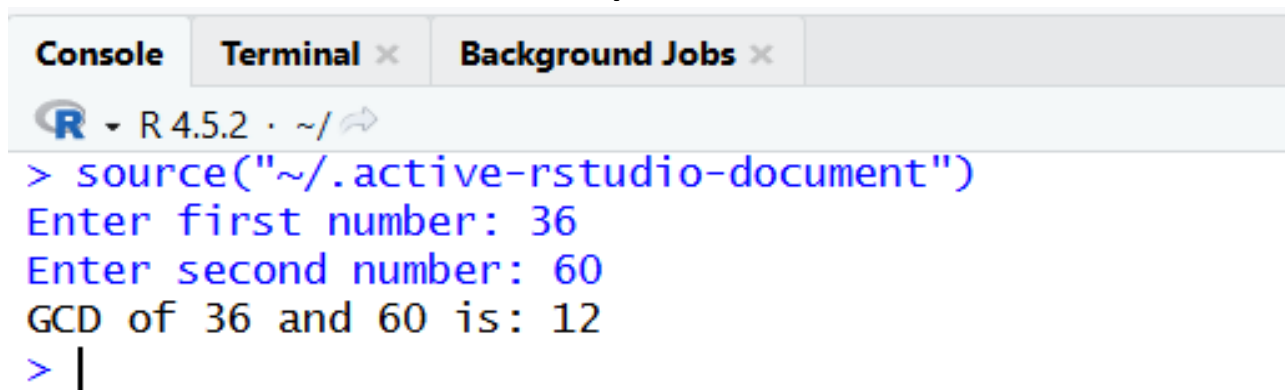
num1 <- as.integer(readline(prompt = "Enter first number: "))
num2 <- as.integer(readline(prompt = "Enter second number: "))

# Finding GCD

result <- gcd(abs(num1), abs(num2))

# Displaying result

cat("GCD of", num1, "and", num2, "is:", result)
```

Output:A screenshot of the RStudio interface showing the Console tab. The console displays the execution of the R script. It starts with the R logo and version 'R 4.5.2 · ~/'. The user enters the command to source the script. The prompt 'Enter first number:' is followed by the input '36'. The prompt 'Enter second number:' is followed by the input '60'. The final output is 'GCD of 36 and 60 is: 12'. The prompt '>' is followed by a vertical bar '|'.

```
Console Terminal × Background Jobs ×
R 4.5.2 · ~/
> source("~/active-rstudio-document")
Enter first number: 36
Enter second number: 60
GCD of 36 and 60 is: 12
> |
```